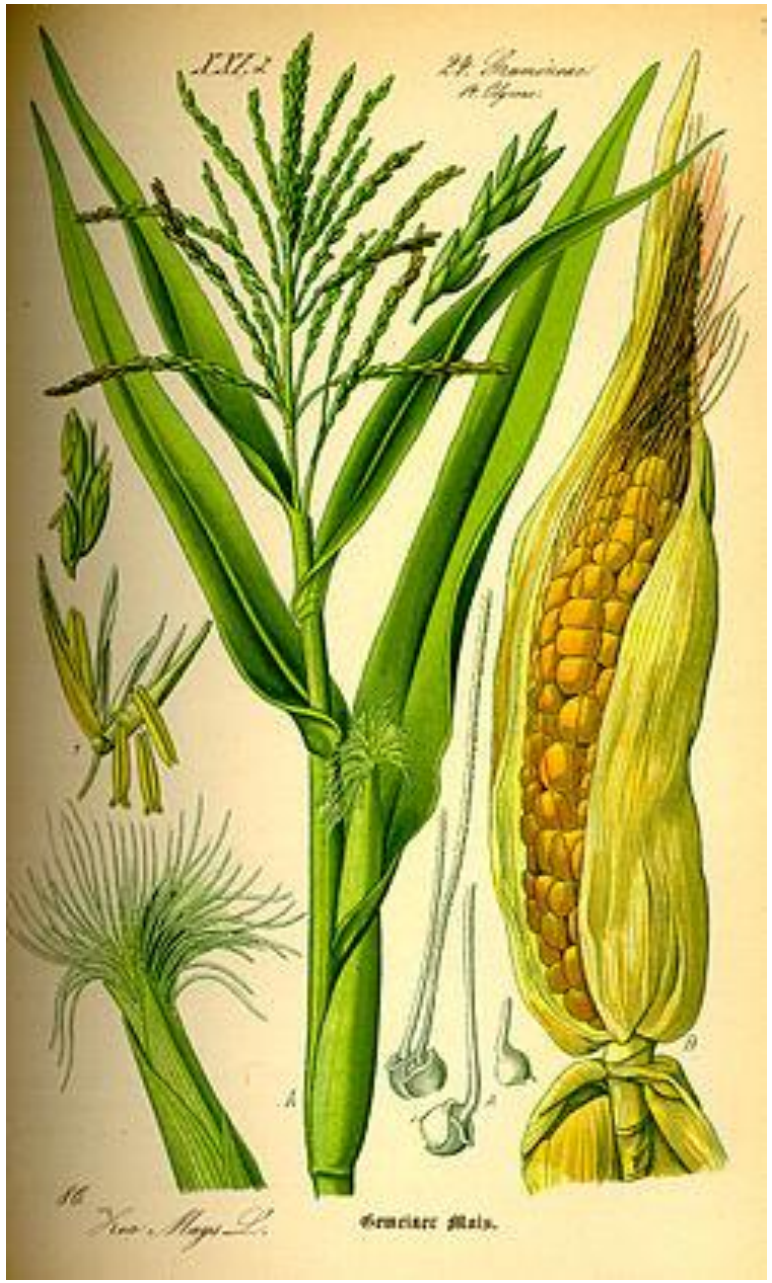


Exploring Knowledge Systems: A Transknowledging Activity

By: Emmanuel Sepúlveda Guzmán

Have you ever been told that you don’t know something? Or that you don’t know enough? Or has your opinion been dismissed with no grounds?

Guiding questions

How does science work?

How is knowledge developed?

sci·ence

/ˈsīəns/

noun

1. **the systematic study of the structure and behavior of the physical and natural world through observation, experimentation, and the testing of theories against the evidence obtained.**

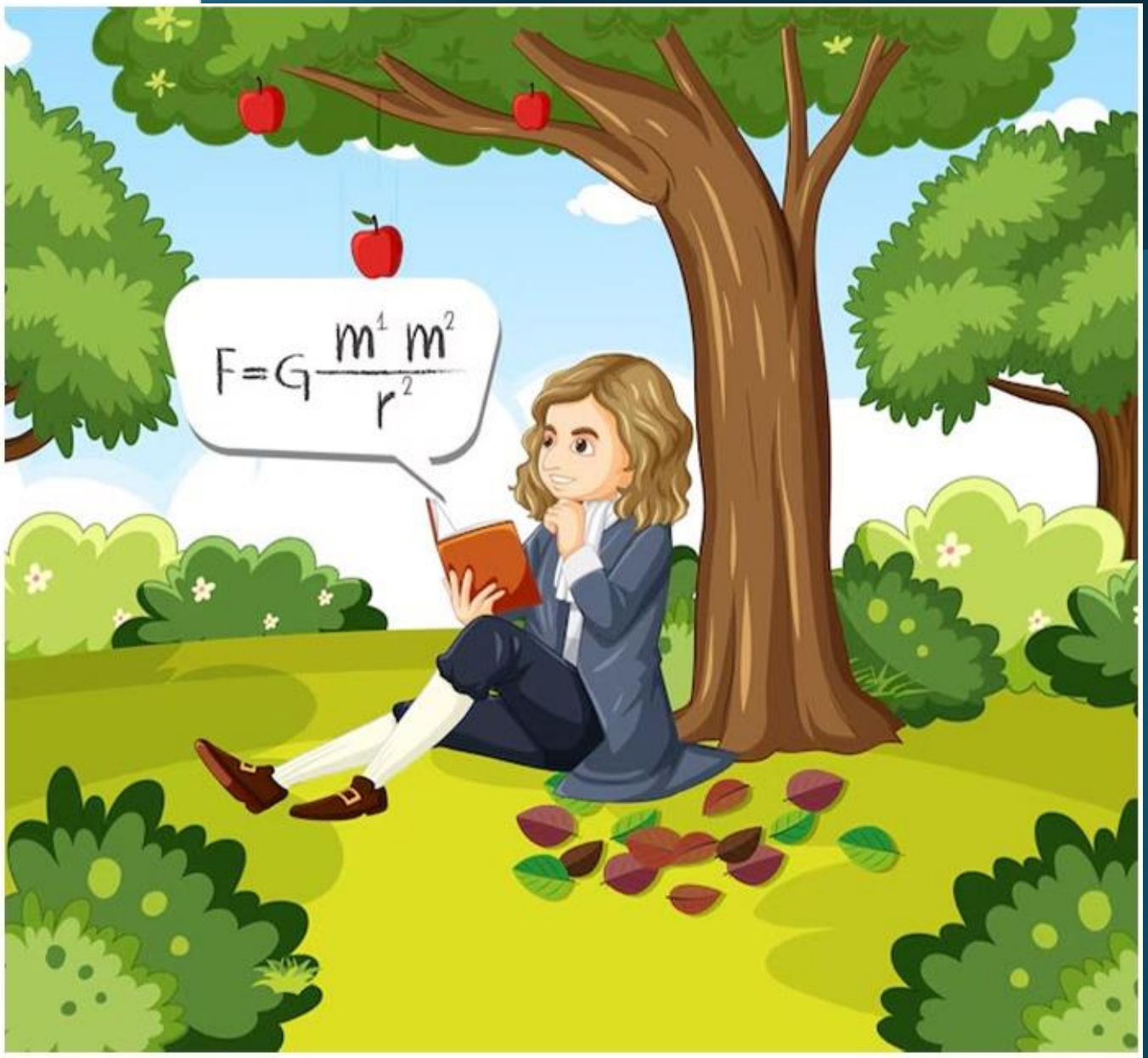
   "the world of science and technology"

   Similar: branch of knowledge | area of study | discipline | field

2. ARCHAIC

   knowledge of any kind.

   "his rare science and his practical skill"


$F=G\frac{m^1 m^2}{r^2}$

Mesquite under stress

Journal of Arid Environments (1992) **23**: 309–319

Influence of water/heat stress on flowering and fruiting of mesquite (*Prosopis glandulosa* var. *glandulosa*)

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The relationships between water stress, inflorescence production, nectar secretion, bee activity and pod production were examined in a self-incompatible, xerophytic tree legume *Prosopis glandulosa* var *glandulosa*. Mature *Prosopis* trees were examined in five study sites located in a 450 to 700 mm annual rainfall gradient over a 2-year period. A drought year produced greater water stress (1·3 MPa) than a wet year (1·0 MPa), had significantly greater inflorescence production and 3·3 times greater pod production. In contrast greater inflorescence lengths and nectar per inflorescence were produced in the wetter year. Inflorescence length was negatively correlated with water stress. Bee visitation was not associated with greater nectar production but it was associated with greater pod production.

Introduction

Over 40 species of xerophytic leguminous trees of the genus *Prosopis* are widely distributed in the semi-arid regions of the world (Simpson, 1977*b*). It is often a major component of native ecosystems of North and South America and *P. juliflora* is one of the few tree species that have been successfully used in the re-forestation of severely degraded ecosystems in Sahelian Africa and India/Pakistan (Felker, 1984). *Prosopis* pods constitute a major resource for desert invertebrates, mammals (Simpson, 1977*b*) and domesticated animals in some of the world's poorest countries (Felker, 1984). These pods were also a major food source for indigenous people in the deserts of the Western Hemisphere (Felger, 1979). Considerable variability exists in the nutrition value of *Prosopis* pods since the protein and sugar content ranges from 9 to 18% and 6 to 41% respectively (Oduol *et al.*, 1986).

Due to the importance of *Prosopis* pod production, both in an ecological and applied context, it is important to examine the influence of water stress on its reproductive biology. Simpson (1977*a*) examined the breeding systems of North and South American *Prosopis*, and insect pollination, but did not examine the influence of water stress on pod production. Phenological observations on *Prosopis glandulosa* in the Sonoran Desert of California revealed high year-to-year variability in pod production (Nilsen *et al.*, 1987; Sharifi *et al.*, 1982). This study found an inverse relationship between precipitation and pod production. These conclusions support the widely-held belief of longtime Texas residents that mesquite produces more pods in drier years.

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3 Easy Ways to Make a Zine

https://www.youtube.com/watch?v=Xs2qlVykwHw

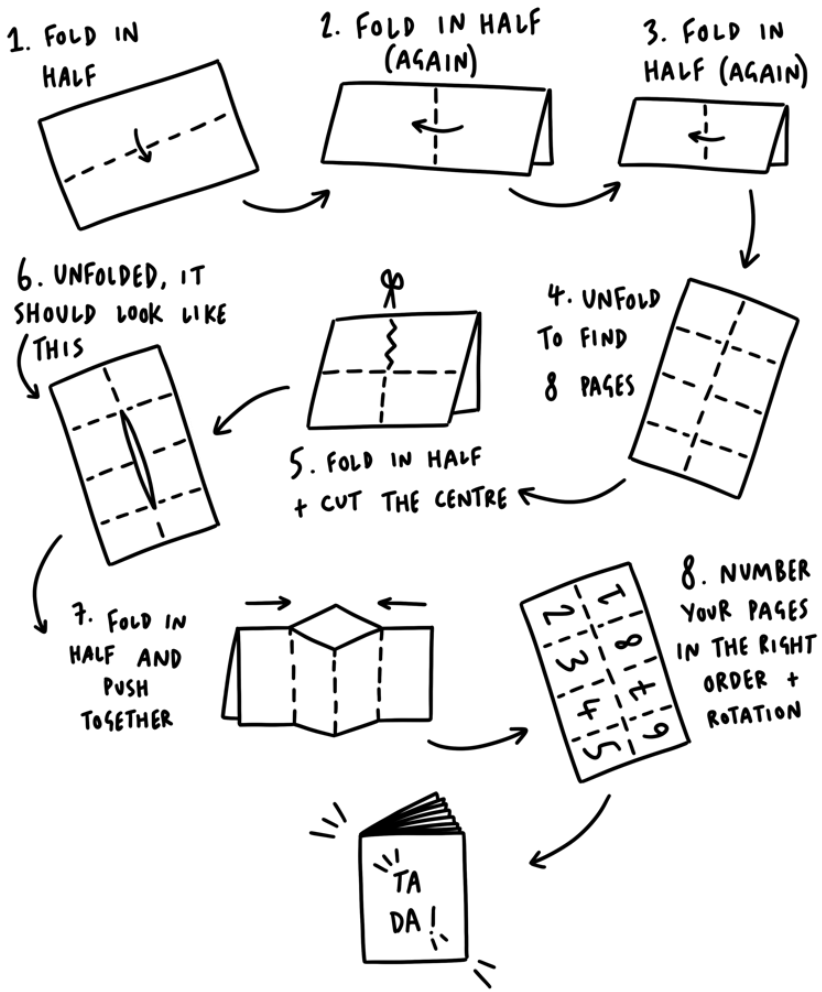
1. Opening Discussion

Divide class into **small groups** of 3-4 students

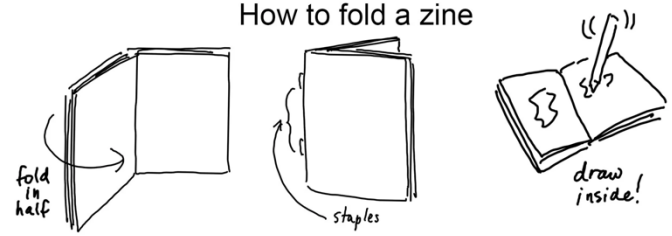
Each group discusses and documents how science is and other ways science might be done according to their understanding of science. Different ways of knowing (e.g., healing practices, agricultural methods, or educational approaches) from their own cultural backgrounds.

2. Produce the plot of your zine

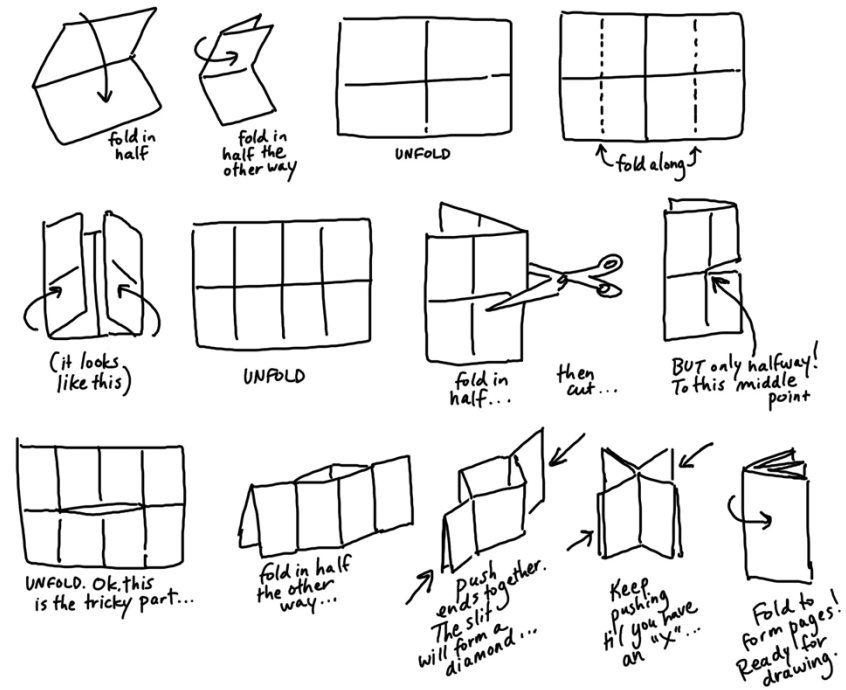
HOW TO MAKE A ZINE



How to fold a zine



This one is good bc it doesn't require staples!



If you unfold the zine, you can make a mini-poster on the other side! I like to use that side as the final panel of my story for MAXIMUM IMPACT!

3. Cross-Cultural Exchange

- Groups rotate to examine other groups' zines
- Add insights and connections
- Identify complementary knowledge systems
- Document instances where different knowledge systems enhance understanding

5. Reflection

- How different knowledge systems complement each other
- Challenges in integrating different ways of knowing
- Personal insights gained about knowledge diversity